Thin-film Compatible Process High Resolution Patterning of Quantum Dots Light-emitting Diodes

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Abstract

Reliable quantum dots (QD) patterning techniques are essential for QD industrialization in the high-resolution display application field. In this work, the patterned quantum dots light-emitting diodes (QLED) are fabricated by a thin film compatible process. SiO_2 is introduced as an isolation layer and the pixel size can shrink to 5 μm with good pattern uniformity. This method shows a promising way to integrate the solution-processed QLED with the traditional semiconductor thin-film process.

Author Keywords

Quantum dots light-emitting diodes; Patterning; Thin-film process

1. Introduction

Quantum dots (QD) based electroluminescent devices are believed to be the next solution for high-quality display.(1,2) The performance of the prototype devices in the laboratory has approached the commercialized standard for a practical display unit.(3–6) However, the compatibility of solution-processed QDs with the current thin-film process is blocking the application of quantum dots light-emitting diodes (QLED).(7) The ink-jet printing is a powerful solution to the obstruction, but still need a bank structure on substrate to isolate the pixel, otherwise, the resolution is seriously limited by the hydromechanics and positioning accuracy of the printer itself.(8) The requirement of the bank structure for QLED is much thinner than the traditional bank for photoluminescence (PL) application. Here we demonstrate a thin film compatible process to fabricate a patterned substrate with controlled pixel isolated layer. It can be integrated with ink-jet printing to realize high-resolution QLED pixel arrays.

2. Experimental

The QLED is fabricated on a patterned glass substrate by thin-film process. The thin-film process including photolithography, deposition, lift-off and dicing. These processes are finished at the SUSTech Core Research Facilities. Instruments used in this process include Asher, Mask aligner (Karl Suss MA6), Wet Bench, Step profiler (KLA P-7), Dicing system. The QDs and ZnMgO nanocrystals in this work is from Shenzhen Planck Innovation Technologies Co. Ltd. Photo resist is RZJ304 which thickness is ~4 μm at 1500 rpm/min. The developer is RZX3038.

The standard photolithography process including substrate treatment, spin coating photo resist, pre-bake, expose, develop and hard bake is carried out according to the datasheet of the photoresist.

Subsequently, both the inverted and normal structure QLED are fabricated on the patterned substrate. The inverted structure is Al/HAT-CN/NPB/TCTA/QD/ZnMgO/ITO and the normal structure is Al/ZnMgO/QD/TFB/PEDOT:PSS/ITO. The TCTA, NPB, HAT-CN layer are evaporated and ZnMgO, TFB, PEDOT:PSS layer are formed by spin coating. The QD layers are formed by spin coating and ink-jet printing, separately. These two methods can demonstrate EL pattern image in good quality and uniformity. The detailed process and parameters are reported in our previous work.(8)

3. Results and Discussion

The final structure of the patterned substrate is shown in Figure 1 step 8. The patterned SiO_2 (P-SiO_2) is deposited on the patterned ITO (P-ITO) electrode as pixel isolation. This structure is fabricated by traditional semiconductor thin-film process, including thin-film deposition and photolithography. All pattern is transferred to the film by lift-off which also can be done by etching. Here the lift-off process is demonstrated. It contains twice photolithography (Step 2, Step 6) and twice lift-off (Step 6, Step 8) steps to define ITO electrode pattern and SiO_2 isolation layer pattern in the whole process. During the process, the ITO and SiO_2 layer is deposited by sputtering (Step 3, Step 7). The higher energy accompany by the deposition can degenerate the photoresist, so the thickness of the photoresist (RZJ304) is selected to 4 μm the maximum value of this photoresist in order to reduce the lift-off difficulty.

In fact, the etching process is more suitable for patterned substrate fabrication due to the resolution of the lift-off process is limited. Especially the SiO_2 isolation layer can be deposition by E-beam Evaporation or LPCVD. This process is compatible with the traditional semiconductor process in display unit, the gate dielectrics of thin-film transistor both bottom gate and top gate can be fabricated with this isolation layer simultaneously. This also benefits cost control and yield.

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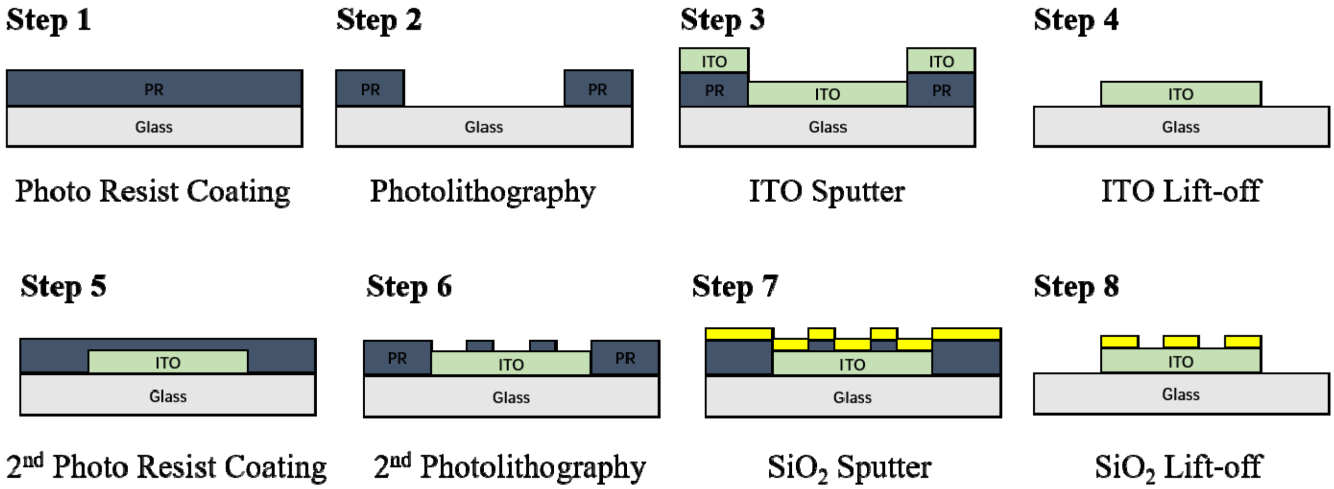

Figure 1. Process flow of patterning substrate for QLED pixel fabrication in this work in which SiO_2 layer is introduced to isolate the pixel. An alternate flow is using LPCVD SiO_2 and ICP etching to form pixel patterns which can get higher resolution and isolation layer thickness in principle.

In the process shown in Figure 1, both the macroscopic (1 cm × 1 cm) and microscopic (5 μm × 5 μm) patterns are fabricated. Figure 2 shows the QLED device with a macroscopic pattern. The active area is defined not only by the overlay of the anode and the cathode, but also the isolation layer. Figure 2 (a) shows the designed mask layout. The ITO and alignment mark is predefined (left) so the active area defined by SiO_2 pattern (right) can be aligned properly. Figure 2 (b) is a photo of patterned substrate the size of the pattern is 1 cm x 1 cm. After dicing, the patterned substrate is transferred to the QLED fabrication process. Here the inverted structure Al/HAT-CN/NPB/TCTA/QD/ZnMgO/P-SiO_2/ITO is employed, in which the ZnMgO and QD are spin coated and other layers are deposited by thermo evaporation. Figure 2 (c) is the photo-luminescence image under 365 nm UV excitation. The pattern is defined with high quality. The EL images of the device are shown in Figure 2 (d) and (e). The details of logo are clearly shown.

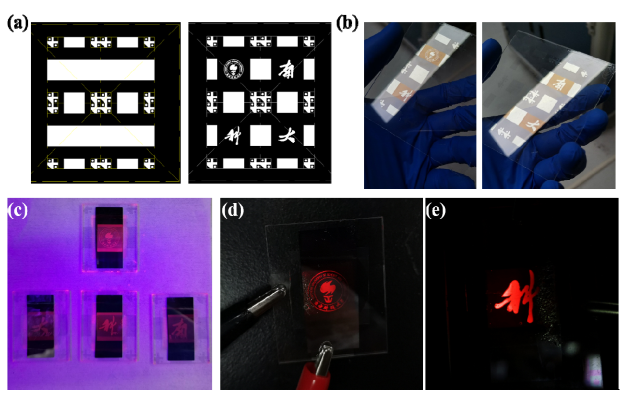

Figure 2. EL devices with 1 cm × 1 cm pattern. **(a)** Designed layout of ITO and SiO_2 Pattern. **(b)** Photo of patterned substrate in Step 8 (P-SiO_2/P-ITO/Glass). **(c)** the image of the inverted device (TCTA/NPB/HAT-CN as HTL and ZnMgO NCs as ETL) under 365 nm UV excitation. **(d)** and **(e)** are the EL images of patterned QLED, the size of glass substrate is 26 mm × 32 mm the active area is 1 cm × 1 cm.

The predefined patterns can be narrowed tom5 μm × 5 μm arrays. Figure 3 (a) (b) shows an optical image of the P-SiO_2/P-ITO/Glass substrate with circle of 20 μm in diameter and square with 5 μm side length, respectively. Figure 3 (c) is the device architecture for patterned QLED in this work. For microscopic pattern, the normal structure QLED is employed. The EL image Figure 3 (d) and (e) shows the patterned QLED with outstanding pattern quality and uniformity. The resolution and accuracy are apparently improved compared with all inkjet printed device. It worth noting that the ink-jet printing QLED is also fabricated on the patterned substrate shown in Figure 3 (f). The optimized QD ink in our previous work is utilized to suppress the coffee ring effect. It shows a potential application of this thin-film compatible process patterned substrate with other patterning technology in high-resolution QLED display.

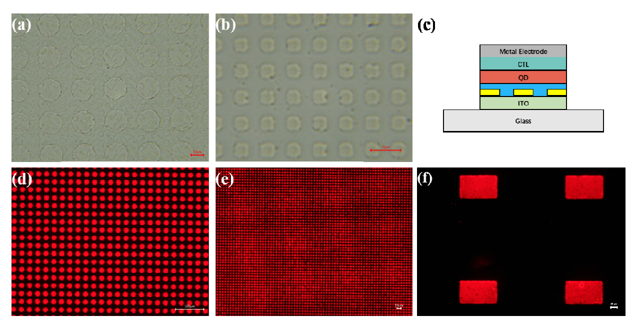

Figure 3. EL devices with micron-level size pattern. **(a)** Patterned substrate (P-SiO_2/P-ITO/Glass) with 20 μm diameter circle under optical microscope. **(b)** Patterned substrate (P-SiO_2/P-ITO/Glass) with 5 μm square under optical microscope. **(c)** Device architecture of red QLED based on the patterned substrate with SiO_2 isolation layer. **(d)** EL image of QLED with 20 μm diameter circle pattern. **(e)** EL image of QLED with 5 μm square pattern. **(f)** EL image of ink-jet printing QLED with hundred micrometer rectangle pattern using optimized QD ink (our previous work).

4. Conclusion

The patterned substrate with SiO_2 as an isolation layer is fabricated with thin-film compatible process. With an optimized process parameter, the high resolutional QLED pixel array of 5 μm × 5 μm size pattern is demonstrated in good quality and uniformity. The primary advantages of this technique are listed below:

- High resolution and good process control.
- Thin-film process compatible.
- Easy to integrated with thin-film transistor.
- Easy to integrated with ink-jet printing process.

Since the mature of the semiconductor technology, this method shows great potential in mass production of QLED and good integration with other counterparts in display unit.

5. Acknowledgements

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6. References

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